worlds calling out to worlds—while we on Earth are alive at the critical moment when we first decide to listen?

Our species has discovered a way to communicate through the dark, to transcend immense distances. No means of communication is faster or cheaper or reaches out farther. It's called radio.

After billions of years of biological evolution—on their planet and ours—an alien civilization cannot be in technological lockstep with us. There have been humans for more than twenty thousand centuries, but we've had radio only for about one century. If alien civilizations are behind us, they're likely to be too far behind to have radio. And if they're ahead of us, they're likely to be far ahead of us. Think of the technical advances on our world over just the last few centuries. What is for us technologically difficult or impossible, what might seem to us like magic, might for them be trivially easy. They might use other, very advanced means to communicate with their peers, but they would know about radio as an approach to newly emerging civilizations. Even with no more than our level of technology at the transmitting and receiving ends, we could communicate today across much of the Galaxy. They should be able to do much better.

If they exist.

But our fear of the dark rebels. The idea of alien beings troubles us. We conjure up objections:

"It's too expensive." But, in its fullest modern technological expression, it costs less than one attack helicopter a year.

"We'll never understand what they're saying." But, because the message is transmitted by radio, we and they must have radio physics, radio astronomy, and radio technology in common. The laws of Nature are the same everywhere; so science itself provides a means and language of communication even between very different kinds of beings—provided they both have science. Figuring out the message, if we're fortunate enough to receive one, may be much easier than acquiring it.

"It would be demoralizing to learn that our science is primitive." But by the standards of the next few centuries, at least some of our present science will be considered primitive, extraterrestrials or no